

SQL Notes

SQL is Structured Query Language.

SQL database management system (MySQL,PostgreSQL,SQLite,Oracle,DB2). These are like software where SQL language can be executed.

It is a language used to get the data from database.

Data is stored in a table. Each row is a record and column is a property.

Students

S. No.	Name	Last_name	Age	Marks
1	Roy	Joe	34	90
2	Lena	Dzousa	30	75
3	Ajay	Pathak	45	87
4	Ezaam	Joe	24	65
5	Devansh	Naik	48	76

SQL Commands

1. Select command: It is used to fetch the data from the table.
`select name, age from Students;`
2. Where condition: It returns only data that match with the given condition
`select name from Students where Marks > 65;`
3. Multiple conditions:
AND: It returns only data when both conditions are true.
`select name from Student where Marks > 65 and Marks < 90;`
OR: It returns data only if at least one condition is true.
`select name from Student where Marks>65 or Marks<90;`
4. LIKE operator:
It is used for searching text patterns.
Starts with 'A'
`select name from Student where name like 'A%';`
Ends with 'A'
`select name from Student where name like '%A';`
Contains 'A'
`select name from Student where name like '%A%';`
5. Order By:
It is used to sort the results.
Ascending Order
`select name from Student order by salary;`
Descending Order
`select name from Student order by salary desc;`

SQL generally follows the below given Execution order:

- FROM
- WHERE
- SELECT

- ORDER BY
6. Limit:
It restricts the result returned.
`select name from Student limit 4;`

Filtering

1. Between
It is used to filter values within a range.
`select name from Student where marks between 75 and 90;`
This includes value including 30 and 45
It works similar to marks >= 75 and marks<=90;
2. IN
It is used to fetch values that includes multiple exact values.
`select name from Student where last_name in ('Joe' , 'Pathak');`
This works similar to where name = 'Joe' or name = 'Pathak';
3. Not like
It is used like a negative condition.
`select name from student where last_name not like 'J%';`

**`select name from Student
where marks
between 65 and 89
and last name not like 'J%'
order by name desc
limit 2;`**

Aggregate Functions

It summarizes the data.

1. Count()
This counts the rows.
`select count(*) from Student;`
2. Average()
This calculates the average.
`select avg(marks) from Student;`
3. Sum()
This calculates the sum.
`select sum(marks) from Students;`
4. Max()
This gives the maximum value.
`select max(marks) from Students;`
5. Min()
This gives the minimum value.
`select min(marks) from Students;`
6. Group By
 - It groups rows with same value.
 - Aggregate functions work on each group.

- Without Group By, SQL cannot calculate the summaries of a group.

Orders

Item_Id	Items	Amount
1	Printer	2000
2	Keyboard	5000
3	Mouse	600
4	Mousepad	300
5	Keyboard	2000
6	Mouse	790

```
select Items, sum(amount) from Orders group by Items;
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```
select *  
from table_name  
where condition  
group by column  
order by column desc  
limit 5;
```